# DarshanEase - Temple Darshan Ticket Booking System

Project Documentation

## 1. Introduction:

### 1.1 Project Title

**DarshanEase – Online Temple Darshan Ticket Booking System**

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### 1.2 Introduction

In today's digital era, online services have significantly transformed the way people perform their daily activities. From banking and shopping to education and healthcare, technology has simplified many traditional processes. Similarly, religious institutions are also adopting digital solutions to improve the experience of devotees. One such transformation is the introduction of online temple darshan ticket booking systems, which allow devotees to reserve darshan slots in advance without waiting in long queues.

**DarshanEase** is a web-based temple darshan ticket booking application developed using the **MERN Stack**, which consists of **MongoDB**, **Express.js**, **React.js**, and **Node.js**. The application is designed to provide devotees with a secure, efficient, and user-friendly platform for booking darshan tickets online. Instead of visiting temples physically to purchase tickets, users can simply register, log in, browse available temples, select a preferred date and time, and complete the booking process from anywhere using an internet connection.

The application aims to eliminate the inconvenience associated with traditional booking methods. During festivals and special occasions, thousands of devotees visit temples, leading to overcrowding and long waiting hours. DarshanEase solves this problem by introducing an online slot booking system that distributes visitors efficiently throughout the day. This not only improves crowd management but also provides devotees with a peaceful and organized temple visit.

The system also benefits temple authorities by reducing manual work involved in ticket management. Administrators can add new temples, update schedules, manage available slots, monitor bookings, and maintain user information through an easy-to-use administrative dashboard. All booking information is securely stored in the MongoDB database, ensuring data consistency and easy retrieval.

Security is an important aspect of the application. User authentication is implemented using **JSON Web Tokens (JWT)** and passwords are encrypted using **bcrypt** before storing them in the database. This ensures that sensitive user information remains protected against unauthorized access.

DarshanEase also includes additional modules such as online donations, booking history, contact support, and email notifications. These features provide a complete digital solution for temple management while enhancing the overall experience of devotees.

Overall, the project demonstrates how modern web technologies can be effectively used to solve real-world problems. By integrating frontend, backend, database, and authentication technologies into a single application, DarshanEase showcases the practical implementation of full-stack web development concepts.

## 2. Project Overview:

### 2.1 Overview

DarshanEase is a full-stack web application developed using the MERN technology stack. The application is designed to automate the temple darshan booking process by providing an online platform for devotees and temple administrators. It simplifies the reservation process while maintaining transparency, security, and convenience.

The frontend of the application is developed using React.js, which provides an interactive and responsive user interface. The backend is implemented using Node.js and Express.js, which handle API requests, authentication, booking management, and communication with the database. MongoDB serves as the primary database for storing user information, temple details, booking records, and donation information.

The project follows a client-server architecture where the React frontend communicates with the Express backend through REST APIs. The backend processes user requests, validates data, performs authentication, and stores or retrieves information from MongoDB before returning responses to the frontend.

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## 2.2 Scope of the Project

The scope of DarshanEase includes both user and administrator functionalities.

### User Module

- Registration
- Login
- Temple browsing
- Darshan slot selection
- Ticket booking
- Donation
- Booking history
- Contact support

### Administrator Module

- Admin login
- Temple management
- Booking management
- User management
- Slot management
- Donation management
- Dashboard monitoring

## 3. Architecture:

### 3.1 Overview

The DarshanEase application follows a **three-tier architecture**, consisting of the Presentation Layer, Application Layer, and Database Layer. This architecture improves maintainability, scalability, and modularity.

User

|

React Frontend

|

REST APIs

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Express.js + Node.js

MongoDB Database

### 3.2 Frontend Architecture

The frontend is developed using **React.js**. It provides an interactive user interface where users can register, log in, browse temples, book tickets, and manage their bookings.

Main frontend responsibilities include:

- Rendering web pages.
- Form validation.
- State management.
- API communication.
- User navigation.
- Responsive design.

Major frontend folders include:

- Components
- Pages
- Assets
- Services
- Routes

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### 3.3 Backend Architecture

The backend is built using **Node.js** and **Express.js**. It acts as the bridge between the frontend and the database.

Responsibilities include:

- User authentication.
- Booking management.
- Business logic.

- REST API implementation.
- Database communication.
- Error handling.

The backend contains:

- Controllers
- Routes
- Models
- Middleware
- Configuration files

## 4. Setup Instructions:

### 4.1 Prerequisites

Before running the DarshanEase application, ensure that the following software is installed on your system:

| **Software** | **Version** |
|---|---|
| Node.js | v18 or above |
| npm | Latest Version |
| MongoDB Community Server / MongoDB Atlas | Latest |
| Git | Latest |
| Visual Studio Code | Latest |
| Google Chrome | Latest |

**Required Technologies**

- React.js
- Node.js
- Express.js

- MongoDB
- Mongoose
- JWT
- bcrypt
- Nodemailer

### 4.2 Required NPM Packages

Backend Packages

- express
- mongoose
- dotenv
- cors
- jsonwebtoken
- bcryptjs
- nodemailer
- cookie-parser

Frontend Packages

- react
- react-router-dom
- axios
- bootstrap (if used)

## 4.3 Project Configuration:

The backend server connects to MongoDB using Mongoose.

Frontend communicates with backend through REST APIs using Axios.

JWT tokens are stored securely after login to authenticate future requests.

## 5. Folder Structure:

The project follows a modular folder structure, making it easy to maintain and extend.
DARSHAN-main

```
│
├── frontend
│   ├── public
│   ├── src
│   │   ├── assets
│   │   ├── components
│   │   ├── pages
│   │   ├── services
│   │   ├── routes
│   │
│   └── package.json
│
├── backend
│   ├── config
│   ├── controllers
│   ├── middleware
│   ├── utils
│   ├── server.js
│   ├── package.json
│   └── .env
```

## 6. Running the Application:

Once all dependencies are installed and MongoDB is configured, the application can be executed locally.

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### Step 1

Start MongoDB

If using MongoDB Atlas, ensure the cluster is running.

### Step 2

Start Backend

Navigate to backend.

### Step 3

Start Frontend

Navigate to frontend

### Step 4

Open Browser

[http://localhost:5173](http://localhost:5173)

The home page of DarshanEase will be displayed.

## 7. API Documentation:

The backend of DarshanEase exposes several REST APIs that allow communication between the frontend and the server. These APIs perform user authentication, temple management, booking operations, donations, and contact management.

**Temple APIs**

| Method | Endpoint | Description |
|---|---|---|
| GET | /api/temples | View Temples |
| GET | /api/temples/:id | Temple Details |
| POST | /api/temples | Add Temple |
| PUT | /api/temples/:id | Update Temple |
| DELETE | /api/temples/:id | Delete Temple |

| Method | Endpoint | Description |
|---|---|---|
| POST | /api/booking | Book Ticket |
| GET | /api/booking | Booking History |
| DELETE | /api/booking/:id | Cancel Booking |

**API Testing**

The APIs were tested using **Postman** to ensure that each endpoint returned the expected response. Both successful and error scenarios were verified, including user registration, login authentication, booking creation, donation submission, and contact form handling. Proper HTTP status codes and JSON responses were validated during testing to confirm the reliability and correctness of the backend services.

# 8. Authentication:

## 8.1 Overview:

Authentication is one of the most important components of the DarshanEase application. It ensures that only authorized users can access specific features such as booking darshan tickets, viewing booking history, and managing temple information. The application implements **JSON Web Token (JWT)** based authentication along with **bcrypt** password hashing to provide a secure login system.

When a new user registers, their password is encrypted using bcrypt before being stored in the MongoDB database. This prevents passwords from being stored in plain text and protects user credentials in case of unauthorized database access. During login, the entered password is compared with the encrypted password stored in the database.

After successful verification, the server generates a JWT token and sends it to the client. The frontend stores this token securely and includes it in the Authorization header for all protected API requests. The backend verifies the token before allowing access to secured resources.

This mechanism ensures that only authenticated users can perform operations such as booking darshan tickets or accessing their booking history, while administrators have separate authorization privileges to manage temples, bookings, and users.

## 8.2 Password Security:

The project uses **bcrypt** to hash passwords before storing them in the database.

Advantages include:

- Passwords are never stored in plain text.
- Protects against database leaks.
- Provides secure password verification.
- Industry-standard password encryption.

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## 8.3 Authorization:

The system supports two types of users:

**User**

- Register account
- Login
- View temples
- Book darshan tickets
- View booking history
- Donate
- Contact temple administration

**Administrator**

- Login
- Manage temples
- Manage users
- View bookings
- Manage darshan slots
- View donations
- Manage contact requests

# 9. User Interface:

## 9.1 Overview:

The user interface of DarshanEase is developed using **React.js** with a focus on simplicity, responsiveness, and ease of navigation. The interface is designed so that users of all age groups can easily browse temples, select darshan slots, and complete bookings without technical difficulty.

The application follows a responsive layout, ensuring proper display on desktops, laptops, tablets, and mobile devices. Navigation between pages is handled using React Router, providing a smooth and seamless user experience without full-page reloads.

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## 9.2 Home Page:

The home page is the first screen displayed to users. It introduces the DarshanEase platform and provides navigation links to login, registration, temples, donations, and contact pages.

**Features:**

- Navigation Bar
- Hero Section
- Featured Temples
- Footer
- Responsive Layout

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## 9.3 Registration Page:

This page allows new users to create an account.

Fields include:

- Name
- Email
- Password
- Confirm Password

The entered information is validated before registration.

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## 9.4 Login Page:

Users enter their email and password to access their account.

Features:

- JWT Authentication
- Error Handling
- Remember Session (if implemented)

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## 9.5 Temple Listing Page:

Displays all available temples with relevant details such as:

- Temple Name
- Location
- Description
- Image
- View Details Button

## 9.6 Booking Page:

Allows users to:

- Select temple
- Choose date
- Select available time slot
- Confirm booking

### 9.7 Booking Confirmation:

After successful booking, users receive a confirmation message containing booking details.

## 9.8 Donation Page:

Users can make voluntary donations to temples through the application.

## 9.9 Contact Page:

Provides a contact form where users can submit feedback, suggestions, or queries.

# 10. Testing:

## 10.1 Overview:

Testing ensures that every feature of the application functions correctly and meets the project requirements. Various testing techniques were applied to verify the functionality, performance, and reliability of the DarshanEase system.

The project was tested during development to identify and fix errors before deployment.

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## 10.2 Types of Testing:

### Functional Testing

Ensures all features work as expected.

Examples:

- User Registration
- Login
- Temple Display
- Booking
- Donation

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### Authentication Testing:

Checks:

- Valid Login
- Invalid Login
- JWT Token Verification
- Protected Routes

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### Database Testing:

Verifies:

- Data Storage
- Data Retrieval
- Record Updates
- Record Deletion

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### API Testing:

REST APIs were tested using Postman.

Verified:

- HTTP Methods
- Status Codes
- JSON Responses
- Authentication

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### Responsive Testing:

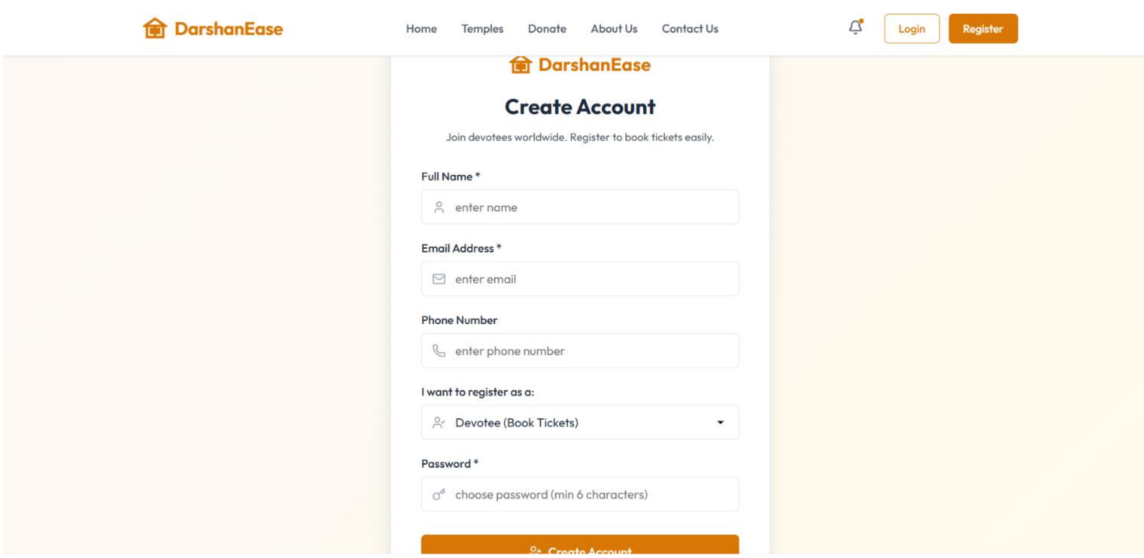
Application tested on:

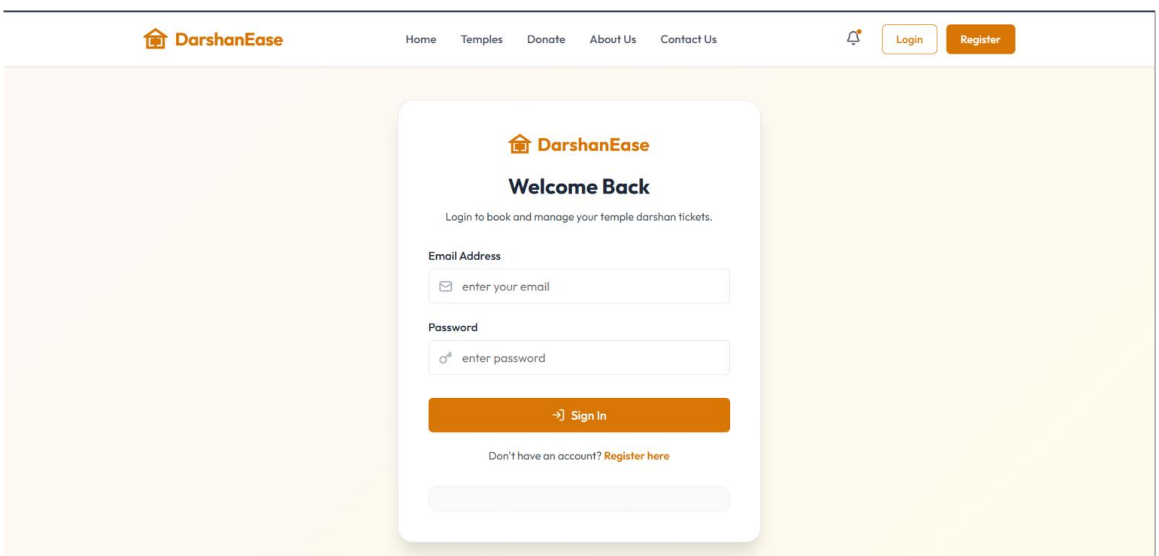
- Desktop
- Laptop
- Tablet
- Mobile

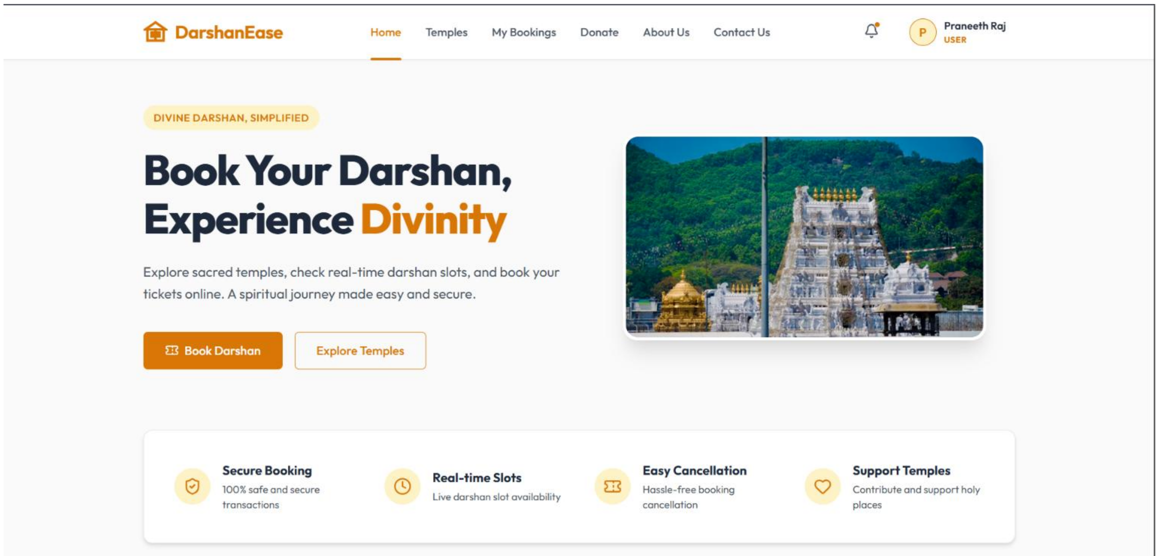
## 10.3 Testing Tools:

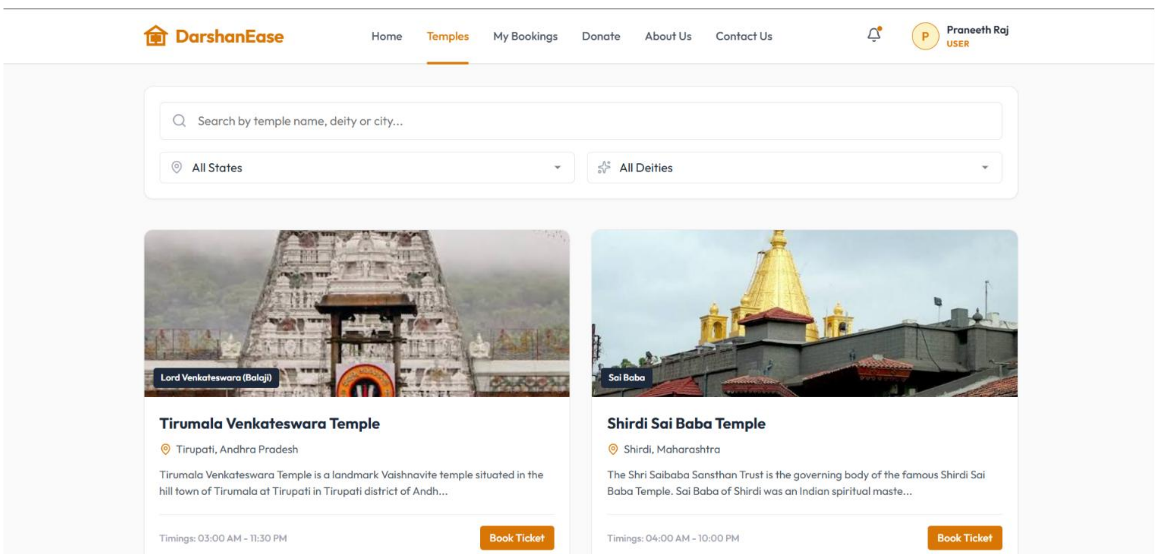
- Postman
- Google Chrome
- MongoDB Compass
- Visual Studio Code

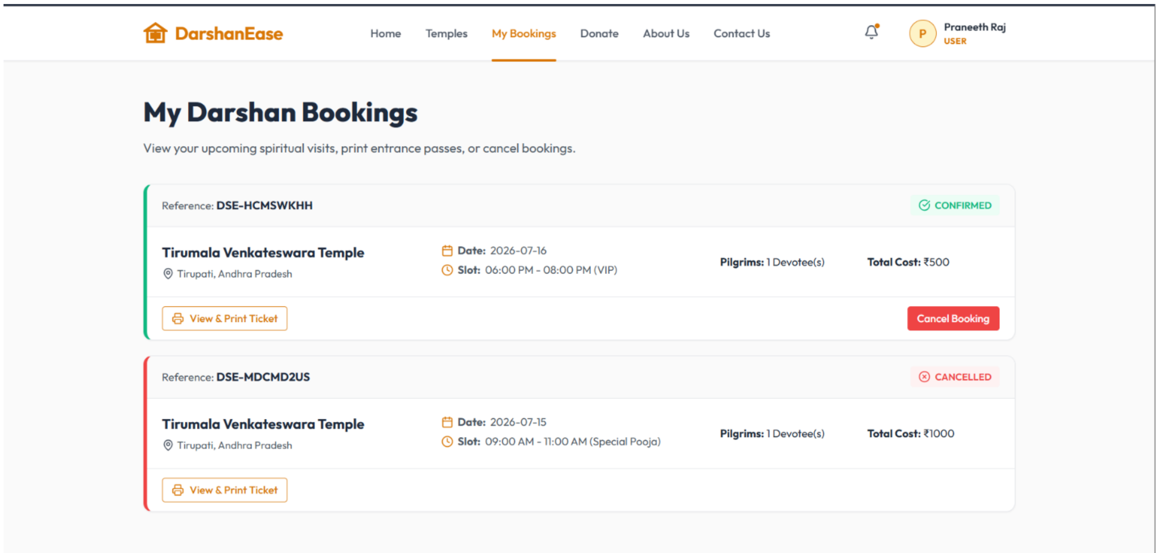
# 11. Screenshots / Demo:




DarshanEase
Home
Temples
My Bookings
Donate
About Us
Contact Us
Praneeth Raj
USER
DIVINE DARSHAN, SIMPLIFIED
Book Your Darshan, Experience Divinity
Explore sacred temples, check real-time darshan slots, and book your tickets online. A spiritual journey made easy and secure.
Book Darshan
Explore Temples
Secure Booking
100% safe and secure transactions
Real-time Slots
Live darshan slot availability
Easy Cancellation
Hassle-free booking cancellation
Support Temples
Contribute and support holy places


DarshanEase
Home
Temples
My Bookings
Donate
About Us
Contact Us
Praneeth Raj
USER
Search by temple name, deity or city...
All States
All Deities
Lord Venkateswara (Balaji)
Tirumala Venkateswara Temple
Tirupati, Andhra Pradesh
Tirumala Venkateswara Temple is a landmark Vaishnavite temple situated in the hill town of Tirumala at Tirupati in Tirupati district of Andh...
Timings: 03:00 AM - 11:30 PM
Book Ticket
Sai Baba
Shirdi Sai Baba Temple
Shirdi, Maharashtra
The Shri Saibaba Sansthan Trust is the governing body of the famous Shirdi Sai Baba Temple. Sai Baba of Shirdi was an Indian spiritual maste...
Timings: 04:00 AM - 10:00 PM
Book Ticket




## 12. Known Issues:

Although the application performs all core functionalities successfully, a few limitations remain.

- Online payment gateway is not integrated.
- SMS notifications are not available.
- QR-code ticket generation is not implemented.
- No real-time seat or slot synchronization for high traffic.
- Multi-language support is not available.
- Internet connection is required for all operations.
- Analytics and reporting features are limited.

These issues do not affect the core booking functionality but present opportunities for future improvements.

## 13. Future Enhancements:

The DarshanEase application can be enhanced with several advanced features to improve user convenience and system capabilities.

## Proposed Enhancements:

- Online Payment Gateway Integration
- QR Code-Based Darshan Tickets
- SMS and WhatsApp Notifications
- Mobile Application (Android & iOS)
- Multi-language Support
- Live Darshan Streaming
- AI-Based Crowd Prediction
- Google Maps Navigation to Temples
- Facial Recognition Entry System
- Chatbot for User Support
- Advanced Analytics Dashboard for Administrators
- Cloud Deployment (AWS, Azure, or Google Cloud)
- Push Notifications
- Digital Receipts for Donations
- Integration with UPI Payment Systems

## Benefits of Future Enhancements:

- Improved user experience.
- Better crowd management.
- Faster ticket verification.
- Enhanced security.
- Greater accessibility.
- Increased scalability.
- Better administrative insights.

## Conclusion:

DarshanEase successfully demonstrates the practical implementation of a full-stack web application using the MERN stack. By digitizing temple darshan ticket booking, the system minimizes manual effort, reduces waiting time, and improves the overall experience for devotees. The application combines a responsive React frontend, a secure Express/Node.js backend, MongoDB for reliable data storage, and JWT-based authentication to deliver a robust and scalable solution. The project not only fulfills its functional objectives but also provides a strong foundation for future enhancements such as online payments, QR-code ticket verification, and mobile application support.